

ANSWER KEY — Integration Lesson 15 Test: Two Special Functions, e^x and $1/x$

For the grader · full working shown · give credit for correct method with small arithmetic slips · accept exact OR correctly rounded decimal forms

1. Evaluate: (a) $\ln 1$ (b) $\ln e$ (c) $\ln(e^4)$, saying what question $\ln$ answers.

Work: $\ln x$ answers "e to WHAT power gives x?" (a) $e^0 = 1$, so $\ln 1 = 0$. (b) $e^1 = e$, so $\ln e = 1$. (c) e to the power 4 gives e^4 , so $\ln(e^4) = 4$. **Answer: (a) 0; (b) 1; (c) 4.**

2. Find $\int e^x dx$ and explain why it's the easiest antiderivative.

Work: e^x is its own derivative, so it is also its own antiderivative — the answer is the function itself, nothing to adjust or divide. **Answer: $e^x + C$.** (Explanation must mention that e^x is its own derivative, in some wording.)

3. Find $\int (6/x) dx$.

Work: $6/x = 6 \times (1/x)$; the constant passes through, and $\int (1/x) dx = \ln x + C$. Check: derivative of $6 \ln x$ is $6/x$ ✓. **Answer: $6 \ln x + C$.**

4. Find $\int (x^3 + 2/x) dx$.

Work: term by term: $\int x^3 dx = x^4/4$ (power rule); $\int (2/x) dx = 2 \ln x$. One $+ C$ covers all. Check: derivative of $x^4/4 + 2 \ln x$ is $x^3 + 2/x$ ✓. **Answer: $x^4/4 + 2 \ln x + C$.**

5. Evaluate $\int_1^{e^3} (1/x) dx$.

Work: antiderivative $\ln x$; FTC: $\ln(e^3) - \ln 1 = 3 - 0 = 3$. **Answer: 3.**

6. Evaluate $\int_0^1 4e^x dx$ (exact, then decimal).

Work: antiderivative $4e^x$; FTC: $4e^1 - 4e^0 = 4e - 4 = 4(e - 1)$. Decimal: $4 \times 1.72 \approx 6.87$. **Answer: exact $4(e - 1)$ (equivalently $4e - 4$); decimal ≈ 6.87 .**

7. Why can't the power rule handle $1/x$, and what does $\int (1/x) dx$ equal?

Work: $1/x = x^{-1}$, so the rule would compute $x^0/(-1 + 1) = x^0/0$ — division by zero, which is meaningless. $n = -1$ is the one exponent the rule cannot do. The antiderivative is a different kind of function entirely: **$\int (1/x) dx = \ln x + C$ (for positive x).**

8. The special property of e^x , and its meaning for a population following e^t .

Work: e^x is its OWN derivative — at every point its slope equals its height. For a population of size e^t , the growth rate always equals the current population size: growth proportional to size (more bacteria make more bacteria). **Answer: e^x is its own derivative (height = slope); the population grows at a rate equal to its current size.**

9. Find F with $F'(x) = 1/x$ and $F(1) = 7$.

Work: general antiderivative $F(x) = \ln x + C$. Condition: $F(1) = \ln 1 + C = 0 + C = 7$, so $C = 7$. Check: derivative of $\ln x + 7$ is $1/x$ ✓, $F(1) = 7$ ✓. **Answer: $F(x) = \ln x + 7$.**

10. Evaluate $\int_1^e (3/x + 2x) dx$ (exact, then decimal).

Work: antiderivative: $3 \ln x + x^2$. Top ($x = e$): $3 \ln e + e^2 = 3 + e^2$. Bottom ($x = 1$): $3 \ln 1 + 1 = 0 + 1 = 1$. Exact: $(3 + e^2) - 1 = e^2 + 2$. Decimal: $7.39 + 2 \approx 9.39$. **Answer: exact $e^2 + 2$; decimal ≈ 9.39 .**